MILITARY SPECIFICATION

PNEUMATIC POWER SYSTEMS, HIGH PRESSURE

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general performance, development, compatibility, and verification requirements for an aircraft or missile high pressure pneumatic power system and its components.

1.2 Use. This specification cannot be used for contractual purposes without supplemental information relating to the performance requirements of Pneumatic Power Systems.

1.2.1 Structure. The supplemental information required is identified by blanks within the specification.

1.2.2 Instructional handbook. The instructional handbook, which is contained in the appendix herein, provides the rationale for specified requirements, guidance for inclusion of supplemental information, and a lessons learned repository.

1.3 Deviation. Any projected design for a given application which will result in improvement of system performance (e.g. reduced life-cycle cost or reduced development cost through deviation from this specification) may be incorporated by tailoring the specification. Where the requirements of this specification result in compromise of operational capability, they shall be brought to the attention of the procuring activity for consideration of change.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplemental thereto (if applicable) form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB, OH 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-G-4343 Grease, Pneumatic Systems

STANDARDS

MILITARY

MIL-STD-470 Maintainability Program Requirements (for Systems and Equipments)

MIL-STD-1568 Materials and Processes for Corrosion Prevention and Control in Aerospace Weapon Systems

MIL-STD-1587 Materials and Process Requirements for Air Force Systems

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 System description. The pneumatic power system shall generate and/or store, condition, and distribute pneumatic power to the control and/or actuating devices dependent on pneumatic power for normal, alternate, or emergency operation. The pneumatic power system may include components in using systems dependent upon pneumatic power. The pneumatic power system(s) shall be sized and configured to supply pneumatic power, as required, to the selected using functions in all modes of ground and flight operations. _____ shall be used for guidance.

3.2 Performance requirements

3.2.1 Systems characteristics

3.2.1.1 Materials. Materials selections shall be made in accordance with MIL-STD-1587, and corrosion control shall be established in accordance with MIL-STD-1568, except as modified by _____.

3.2.1.2 Gas medium. The system gas medium shall be _____.

3.2.1.3 System pressure. Main system operating pressure shall be _____ psi at _____ °F. Pressure transients during system operation shall not exceed _____ percent of operating pressure.

3.2.1.3.1 System pressure relief. Overpressure protection shall be provided.

3.2.1.3.2 Back pressure. Proper functioning of any unit shall not be adversely affected by back pressure in the system.

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3.2.1.3.3 System pressure indication. Indicator(s) shall be provided to measure pneumatic pressure.

3.2.1.4 Gas velocity limitations. Tubing size and gas velocity shall be selected to be consistent with design limits and to control actuator rates.

3.2.1.5 Pneumatic power system status indication. The operation of the pneumatic power systems shall be as automatic as practical and the system shall provide, but not be limited to, the following:

a. Information at the _____ which will alert the _____ to unsafe or marginal pneumatic power system operation so that the appropriate corrective actions may be taken.

b. _____.

3.2.1.6 Strength.

3.2.1.6.1 Additional loads. During operation of the aircraft, all pneumatic systems and components which are subjected to structural or other loads which are not of pneumatic system origin shall withstand the combination of loads that can be applied simultaneously in flight. There shall be no system or component failure or permanent set when the specified proof pressure and load combinations are applied at the maximum operating temperature.

3.2.1.6.2 Acceleration loads. Actuation cylinders, other components and their attaching lines and fittings, which in their functional operation are subjected to pressures generated by accelerated loads shall withstand a pressure equal to 100 percent of the maximum pressure that will be developed without failure or permanent set at the maximum operating temperature.

3.2.1.6.3 Proof pressure. No part of a system shall fail, take any permanent set, or be damaged in any manner, when subjected to applicable proof pressure of _____.

3.2.1.6.4 Burst pressure. No component shall rupture when subjected to applicable burst pressure of _____.

3.2.1.7 Environmental conditions. The pneumatic power system equipment shall be capable of withstanding or operating under the following conditions:

Environment	Requirement
Temperature	
Humidity	
Fungus	
Vibration	
Dust	
Salt fog	
Explosive atmosphere	
Acceleration	
Shock	

3.2.1.8 Electromagnetic interference. Electrical components shall meet the following requirements: _____.

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3.2.1.9 Electrical bonding. Pneumatic power system tubing and components shall be electrically bonded in accordance with _____.

3.2.1.10 Fire and explosion hazard proofing. Integration of the pneumatic power system in the vehicle shall preclude the possibility of creating a fire or explosion hazard.

3.2.1.11 Survivability-Vulnerability (S-V). The pneumatic power systems shall be configured to accept a single impact by the threat(s) specified in the System or Air Vehicle Specification without loss of defined mission-essential functions.

3.2.1.11.1 Resistance to gunfire. Pneumatic power system pressure vessels shall not result in fragmentation from a single impact by _____.

3.2.1.11.2 Component clearance. Clearance shall be maintained between moving system components and structure or other components to ensure that no possible combinations of temperature effects, airloads, wear, or structural deflections can cause binding, rubbing, or jamming. There shall also be adequate clearance for component installation, removal, and maintenance.

3.2.2 Functional components characteristics

3.2.2.1 Component selection. Components of the pneumatic power system shall be _____.

3.2.2.2 Actuators. Actuator design and performance requirements shall be defined in the contractor's specification(s). Provision shall be made for control of rates of operation for resisting, assisting, and no-load conditions.

3.2.2.3 Multiple control valves. In systems which incorporate two or more directional control valves, provision shall be made to prevent the medium from being transferred inadvertently at any possible valve setting, from the cylinder ports of one valve into the cylinder ports of another valve.

3.2.2.4 Filters. There shall be means to filter all pneumatic power system gas during flight, ground and servicing operations. Filter elements shall be easily replaced and shall be of the disposable type.

3.2.2.5 Moisture content control. Moisture content shall be limited to _____.

3.2.2.6 Fittings. Straight-threaded fittings shall be used. Unless specifically approved by the procuring activity, no thread lubricants shall be used on straight-threaded fittings. All nonstandard fittings shall require approval by the procuring activity.

3.2.2.7 Packing installation. _____ shall be used as guides for packing installations. Grease conforming to MIL-G-4343 shall be used as a lubricant for Buna N type packings in pneumatic power systems.

3.2.2.8 Seals. Packings and gasket materials shall be compatible with the fluid medium and capable of performing in the specified environment.

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3.2.2.9 Tubing systems

3.2.2.9.1 Tubing systems durability. Tubing and tube connectors shall have a system durability life equal to or greater than the life of the aircraft.

3.2.2.9.2 Tubing identification. All pneumatic lines shall be identified by _____.

3.2.2.9.3 Tubing clearance. Clearance shall be maintained to prevent physical contact of tubing with other tubing, components, or structure as a result of system operation, flight maneuvers, or test pressures. There shall be adequate installation clearance and support for wrenching and removal.

3.2.2.9.4 Repair and replacement of tubing. Repair and replacement methods for tubing and fittings shall be established. This information shall be included in appropriate technical manuals and/or technical orders.

3.2.2.10 Relative motion. Means shall be provided to accommodate relative motion.

3.2.2.11 Air compressor(s). When an airborne compressor package(s) is used, general and specific requirements shall be defined in a Critical Item Development Specification.

3.2.2.12 Air bottles. When air bottles are used, the general and specific requirements, quality assurance provisions, and preparation for delivery requirements shall be defined in a Critical Item Development Specification.

3.2.2.13 Electric-motor-driven compressor. Electric-motor-driven equipment shall be capable of continuous operation.

3.2.2.14 Shelf life. All pneumatic power system components, when properly prepared for storage shall have a minimum shelf life of _____.

3.2.2.15 Endurance. The components of the pneumatic power system shall have a total minimum operating life equal to _____ except for: _____. Endurance cycling life of components shall be established.

3.3 Reliability. The pneumatic power system reliability requirements and allocations shall be derived from the overall system reliability program.

3.4 Maintainability. The pneumatic power system maintainability requirements shall be integrated with the overall system maintainability programs and shall comply with the requirements of MIL-STD-470. Special maintainability requirements are as follows: _____.

3.4.1 Service and access. All pneumatic power system components shall be accessible for inspection, cleaning, adjustment, or replacement, with tools normally found in a mechanic's tool box, while installed and without removal of the engine or major parts of the structure. Exceptions shall be identified.

3.5 System safety. The pneumatic power system safety requirements shall comply with the requirements of _____.

3.6 Interface requirements. Pneumatic power system interfaces shall be identified and controlled to ensure form, fit, and functional compatibility.

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3.6.1 Electrical power system interface. All electrical components shall operate with electrical power defined by _____.

3.6.2 Support equipment (SE) interface. The pneumatic power system shall be compatible with the following SE:

a. _____.

b. _____.

c. _____.

3.6.3 Instrumentation interface(s). Instrumentation interface(s) shall be identified.

4. VERIFICATIONS

4.1 General. The requirements specified in section 4 shall verify the ability of the Pneumatic Power Systems to meet the requirements specified in section 3 herein. Verification shall encompass planning, procedure preparation, and reporting, and shall be accomplished by inspection, analysis, demonstration, or test, or a combination of these methods. All verifications shall be the responsibility of the contractor; the Government reserves the right to witness, perform, any of the verifications.

4.2 Verification of performance requirements

4.2.1 Verification of system characteristics

4.2.1.1 Verification of materials. Materials selection and corrosion control shall be verified as follows: _____.

4.2.1.2 Verification of gas medium. System gas medium selection shall be verified by analysis and inspection.

4.2.1.3 Verification of system pressure. Compliance with system pressure requirements shall be verified by _____.

4.2.1.3.1 Verification of system pressure relief. Pressure relief shall be verified by _____.

4.2.1.3.2 Verification of back pressure. Operating effects from back pressures shall be verified by _____.

4.2.1.3.3 Verification of system pressure indication. Visibility and accuracy of system pressure shall be verified by _____.

4.2.1.4 Verification of gas velocity limitations. The operating characteristics dependent on gas velocities shall be verified by _____.

4.2.1.5 Verification of pneumatic power system status indication. Control and operating features of the pneumatic power system(s) shall be verified by _____.

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4.2.1.6 Strength verification

4.2.1.6.1 Verification of additional loads. The ability to withstand additional loads shall be verified by _____.

4.2.1.6.2 Verification of acceleration loads. The ability of components to withstand pressures which are generated by aceleration loads shall be verified by _____.

4.2.1.6.3 Verification of proof pressure. Proof pressure compliance shall be verified as follows: _____.

4.2.1.6.4 Burst pressure verification. Burst pressure compliance shall be verified by _____.

4.2.1.7 Verification of environmental conditions. Environmental testing shall be conducted as follows: _____.

4.2.1.8 Verification of electromagnetic interference (EMI). Compliance with EMI resistance requirements shall be verified by _____.

4.2.1.9 Verification of electrical bonding. Compliance with electrical bonding requirements shall be verified by inspection and test.

4.2.1.10 Verification of fire and explosion hazard proofing. Fire and explosion resistance shall be verified by _____.

4.2.1.11 Verification of survivability-vulnerability (S-V). S-V of the pneumatic power system shall be verified as follows: _____.

4.2.1.11.1 Verification of resistance to gunfire. Pressure vessel resistance to fragmentation shall be verified as follows: _____.

4.2.1.11.2 Verification of component clearance. Clearance(s) requirements shall be verified by _____.

4.2.2 Verification of functional components characteristics

4.2.2.1 Verification of Component selection. Selection of components shall be verified by _____.

4.2.2.2 Verification of actuators. Actuator performance shall be verified by _____.

4.2.2.3 Verification of multiple control valve(s) operation. The integrity of multiple control valve operation shall be verified by _____.

4.2.2.4 Verification of filters. The required use and effectiveness of filters shall be verified by _____.

4.2.2.5 Verification of moisture content control. Moisture removal effectiveness shall be verified by _____.

4.2.2.6 Verification of fittings. Fitting selection shall be verified by _____.

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4.2.2.7 Verification of packing installation. Packing installation design shall be verified by _____.

4.2.2.8 Verification of seals. Seal compatibility and compliance with leakage requirements shall be verified by _____.

4.2.2.9 Verification of tubing systems

4.2.2.9.1 Verification of tubing systems durability. The durability of tubing systems shall be verified by _____.

4.2.2.9.2 Verification of tubing identification. Tubing identification shall be verified by _____.

4.2.2.9.3 Verification of tubing clearance. Tubing clearance shall be verified by _____.

4.2.2.9.4 Verification of repair and replacement of tubing. Methods for repair and replacement of tubing and fittings shall be verified by _____.

4.2.2.10 Verification of relative motion. Means to provide relative motion shall be verified by _____.

4.2.2.11 Verification of air compressor(s). Verification of the air compressor package shall be by inspection, demonstration, and tests.

4.2.2.12 Verification of air bottles. Air bottle design and performance shall be verified by _____.

4.2.2.13 Verification of electric-motor-driven compressors. Performance of the electric motor-driven equipment shall be verified by component and subsystem tests.

4.2.2.14 Verification of shelf life. Component shelf life capability shall be verified by _____.

4.2.2.15 Verification of endurance. System and component operating life shall be verified by _____.

4.3 Verification of reliability. Compliance with specified reliability requirements shall be verified by _____.

4.4 Verification of maintainability. Compliance with specified maintainability requirements shall be verified by _____.

4.4.1 Verification of service and access. Service and access shall be verified by inspection and demonstration.

4.5 Verification of system safety. Compliance with specified system safety requirements shall be verified by _____.

4.6 Verification of interface requirements. Pneumatic power system interface(s) with other air vehicle systems shall be verified by _____.

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4.6.1 Verification of electrical power system interface. Electrical power system interface shall be verified by _____.

4.6.2 Verification of support equipment (SE) interface. The interface of the pneumatic power system with specified support equipment shall be verified by _____.

4.6.3 Verification of instrumentation interface(s). The presence of required instrumentation shall be verified by _____. The accuracy and range of the instrumentation shall be verified by _____.

5. PACKAGING

5.1 All deliverable items shall be prepared for shipment as directed by the procuring activity.

5.2 Shipment of reservoirs. The shipment of pressurized gas reservoirs by common carrier shall be in compliance with existing regulations of the Department of Transportation (DOT), Hazardous Materials Regulations Board. This Board regulates the transportation of potentially explosive devices for safety reasons. If stored gas bottles are shipped with electro-explosive devices (EEDs) attached, this could involve additional DOT regulations. It is the responsibility of the prime contractor to ensure that the gas bottle supplier has necessary approval from the DOT to ship pressurized gas bottles.

6. NOTES

6.1 Responsible engineering office. The Responsible Engineering Office (REO) for the technical maintenance of this document is ASD/ENFEM. Requests for additional information or assistance can be obtained from Mr. Frank R. Straus, ASD/ENFEM, Wright-Patterson AFB OH 45433; AUTOVON 785-4158, commercial (513) 255-4158. Any information relating to Government contracts must be obtained through contracting officers.

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Preparing Activity
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(Project 1650-F129)

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APPENDIX

PNEUMATIC POWER SYSTEM, HIGH PRESSURE
HANDBOOK FOR

10. SCOPE

10.1 Scope. This appendix provides rationale, guidance, and lessons learned necessary to tailor sections 3 and 4 of the basic specification (MIL-P-87210) for a specific application.

10.2 Purpose. This appendix provides information to assist the government procuring activity in the use of MIL-P-87210.

10.3 Use. This appendix is designed to assist the project engineer in tailoring MIL-P-87210. The blanks of the basic specification shall be filled in to meet operational needs of the equipment being developed.

10.4 Format

10.4.1 Requirement/verification identity. Section 30 of this appendix parallels sections 3 and 4 of the basic specification; paragraph titles and numbering are in the same sequence. Section 30 provides each requirement (section 3) and associated verification (section 4) as stated in the basic specification. Both the requirement and verification have sections for rationale, guidance, and lessons learned.

10.4.2 Requirement/verification package. Section 30 of this appendix has been so arranged that the requirement and associated verification is a complete package to permit addition to, or deletion from the criteria as a single requirement. A requirement is not specified without associated verification.

10.5 Responsible engineering office. The responsible engineering office (REO) for the technical maintenance of the basic specification and this appendix is ASD/ENFEM, Wright-Patterson AFB OH 45433. Requests for additional information or assistance on these documents should be directed to: Mr. F. R. Straus, AUTOVON: 785-4158, commercial: (513) 255-4158. Any information relating to Government contracts shall be obtained through the contracting officer.

20. REFERENCE DOCUMENTS

20.1 Unless otherwise indicated, the following documents are referenced solely to provide supplemental technical data:

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20.1.1 Government documents

SPECIFICATIONS

MILITARY

MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems
MIL-P-5510	Packing, Preformed, Straight Thread Tube Fitting Boss
MIL-J-5513	Joints, Hydraulic Swivel
MIL-G-5514	Gland Design; Packings, Hydraulic, General
MIL-P-5518	Pneumatic Systems, Aircraft Design Installation, and Data Requirements for
MIL-T-5522	Test Requirements and Methods for Aircraft Hydraulic and Emergency Pneumatic Systems
MIL-C-6591	Compressor Unit, Aircraft, Electric Motor Driven, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-M-7969	Motor, Alternating Current, 400-Cycle, 115/200 Volt System, Aircraft, General Specification for
MIL-P-8564	Pneumatic System Components, Aeronautical, General Specification for
MIL-M-8609	Motor, DC, 28 Volt System, General Specification for
MIL-R-8573	Reservoirs, Air, Non-Shatterable Steel
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-G-4343	Lubricating Grease, Pneumatic System
MIL-A-83444	Airplane Damage Tolerance Requirements
MIL-P-83461	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 Degrees F (135 Degrees C)

STANDARDS

MILITARY

MIL-STD-461	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference

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MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of

MIL-STD-470 Maintainability Program Requirements (For Systems and Equipments)

MIL-STD-471 Maintainability Demonstration

MIL-STD-490 Specification Practices

MIL-STD-704 Aircraft Electric Power Characteristics

MIL-STD-785 Reliability Program for Systems and Equipment Development and Production

MIL-STD-810 Environmental Test Methods

MIL-STD-826 Electromagnetic Interference Test Requirements and Test Methods

MIL-STD-875 Type Designation System for Aeronautical and Support Equipment

MIL-STD-882 System Safety Program Requirements

MIL-STD-1247 Markings, Functions and Hazard Designations of Hose, Pipe, and Tube Lines for Aircraft, Missile, and Space Systems

MIL-STD-1523 Age Control of Age-Sensitive Elastomeric Material

MIL-STD-1530 Aircraft Structural Integrity Program, Airplane Requirements

MIL-STD-1568 Materials and Processes for Corrosion Resistance and Control in Aerospace Weapons Systems

MIL-STD-1587 Materials and Process Requirements for Air Force Systems

20.1.2 Other publications

SAE ARP 584 Coiled Tubing

SAE ARP 994 Recommended Practice for the Design of Tubing Installations for Aerospace Fluid Power Systems

SAE ARP 1625 Cold Gas Systems, Missile or Projectiles Design, Installation, Tests and Requirements for

30. REQUIREMENTS AND VERIFICATIONS

3.1 System description. The pneumatic power system shall generate and/or store, condition and distribute pneumatic power to the control and/or actuating devices dependent on pneumatic power for normal, alternate,or emergency operation. The pneumatic power system may include components in using systems dependent upon pneumatic power. The pneumatic power system(s) shall be sized and configured to supply pneumatic power, as required, to the selected using functions in all modes of ground and flight operations. _____ shall be used for guidance.

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REQUIREMENT RATIONALE (3.1)

The function of the pneumatic power system is to deliver gas at sufficient mass flow rates and pressures to actuating devices in specific modes of flight or ground operations. Pneumatic operation is often selected where intermittent use of power is required, and therefore, has been used for aircraft emergency operations e.g. landing gear extension, wheel brakes and wing flaps.

REQUIREMENT GUIDANCE

Many of the requirements for military pneumatic power system design, installation, and general use may be found in the principal documents under section 20 (these documents may also be used as a reference source to prepare specific requirements which are appropriate for particular pneumatic system(s); other inappropriate requirements may be discarded or modified to achieve a "tailored" document). A few principal documents are listed below:

MIL-P-5518 - Pneumatic Systems, Aircraft; Design, Installation, and Data Requirements for.

This specification covers pneumatic system design and installation requirements for airborne-compressor-charged systems and for ground-charged bottle systems.

MIL-R-8573 - Reservoirs, Air, Nonshatterable Steel

This specification covers steel air reservoirs for aircraft and air reservoirs for missile high-pressure pneumatic systems.

MIL-P-8564 - Pneumatic System Components, Aeronautical, General Specification for

This specification covers the general requirements for pneumatic components used in pneumatic systems, conforming to MIL-P-5518, for Type I (-65°F to +160°F) and Type II (-65°F to +275°F).

MIL-T-5522 - Test Requirements and Methods for Aircraft Hydraulic and Emergency Pneumatic Systems

SAE ARP 1625 - Cold Gas Missile Systems

REQUIREMENT LESSONS LEARNED

Pneumatic compressor systems were used in some World War II USAF aircraft after the development of an automatic gun charger and the development of an electric-motor-driven air compressor unit. One of the early military applications was the gun charger for the B-25 medium bomber. The B-29 and B-50 bombers used electric-motor-driven 1500-psi pneumatics to operate bomb bay doors. On a bomb run, the doors could be opened in a fraction of the time compared to electric system operation.

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Some advantages of pneumatic power are:

a. Energy is easily stored because of gas compressibility.

b. Operation is essentially independent of temperature.

c. Distribution lines are relatively small.

d. Fire hazard is negligible.

e. Fast actuation.

Some design concerns are:

a. Not practical for continuous operation e.g. aircraft flight controls.

b. Leakage

c. Moisture removal

d. Lubrication

e. Compressor output at altitude.

f. Dieseling

g. Insufficient actuator column stiffness to resist flutter.

4.1 General. The requirements specified in section 4 shall verify the ability of the Pneumatic Power Systems to meet the requirements specified in section 3 herein. Verification shall encompass planning, procedure preparation, and reporting, and shall be accomplished by inspection, analysis, demonstration, or test, or a combination of these methods. All verifications shall be the responsibility of the contractor; the Government reserves the right to witness, perform, any of the verifications.

VERIFICATION RATIONALE (4.1)

The design and operating characteristics to fulfill the needs of the weapon system must be made known before full-scale development.

VERIFICATION GUIDANCE

The design and operation of the pneumatic power systems may be documented in a contractor data submittal item, "Subsystem Design Analysis Report". The initial report should be submitted prior to Preliminary Design Review (PDR), and Critical Design Review (CDR), with update revisions to keep current with design and analysis changes. The current Data Item Description (DID) number is DI-S-3581 and should be included in the Contract Data Requirements List (CDRL). Operating performance should be shown by appropriate laboratory component and subsystem tests, and applicable ground, taxi and flight tests (using MIL-T-5522 for guidance). Component tests should be performed using applicable military or contractor-prepared specifications. Pass criteria is

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based on successfully meeting all test objectives. Exceptions can be allowed where verification can be substantiated by valid comparisons to systems previously qualified on other programs. Typical verification methods are defined as follows:

a. Inspection. Inspection is defined as visual verification that the system, including system documentation, conform to the specification requirements. Visual verification can be in the form of inspection of the physical installation, and/or inspection of drawings showing physical relationships, or review of documents reflecting qualification status with respect to specification requirements.

b. Analysis. Analysis is defined as verification that specification requirements have been achieved by evaluation of equations, charts, and reduced data, and by comparisons of analytical predictions with available test data, etc. Verification analysis, however, does not include the normal analysis of data generated during ground or flight testing.

c. Demonstration. Demonstration is defined as an uninstrumented test where success is determined by observation only, such as fit and function checks and tests that require simple quantitative measurements.

d. Test. Test is defined as verification of the specification requirements through the application of established test procedures within specified environmental conditions and subsequent compliance confirmation through analysis of the data generated.

e. Verification by similarity. When verification by similarity is proposed, it shall be accomplished by using verification data from previously developed and qualified items. The verification data from the earlier verification shall be submitted with design data to substantiate that:

(1) The equipment is to perform a similar function in the new application as it did in its earlier verification.

(2) The environment and operating limits shall be no more demanding or degrading than in the earlier operation.

(3) The new item does not incorporate differences that would invalidate the criteria of (1) or (2).

(4) The equipment meets requirements of its earlier application(s) as indicated by its MTBF and other field failure data.

Considerable effort is expended when similarity claims are made late in a program and rejected by the procuring activity, or additional testing is required. Similarity claims must be established early in a program.

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3.2 Performance requirements

3.2.1 System characteristics

3.2.1.1 Materials. Materials selections shall be made in accordance with MIL-STD-1587, and corrosion control shall be established in accordance with MIL-STD-1568, except as modified by _____.

REQUIREMENT RATIONALE (3.2.1.1)

Proper selection and application of materials and corrosion control provisions are necessary to assure the long-range reliability, maintainability, and performance of the system, and to take full advantage of Air Force experience.

REQUIREMENT GUIDANCE

Criteria and controls of material selection and application are contained in the basic military standards which are controlled by the Air Force Materials Laboratory. These documents reflect the industry state of the art, and establish standard handling and production procedures.

REQUIREMENT LESSONS LEARNED

4.2.1.1 Verification of materials. Materials selections and corrosion control shall be verified as follows: _____.

VERIFICATION RATIONALE (4.2.1.1)

Materials with unknown or questionable properties should not be used. In addition, there can be a loss of material and function caused by corrosion.

VERIFICATION GUIDANCE

Proper material selection and corrosion control may be verified by drawing inspection. In those cases where there are unusual circumstances, or for new materials not addressed in MIL-STD-1587 or MIL-STD-1568, appropriate tests should be conducted to verify the adequacy of the application. Careful review is required during design reviews and other meetings with the contractor and his vendors to assure that proper attention is given to material selection and corrosion control.

VERIFICATION LESSONS LEARNED

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3.2.1.2 Gas medium. The system gas medium shall be _____.

REQUIREMENT RATIONALE (3.2.1.2)

The selection of the gas medium can be a critical item from the viewpoint of availability, cost, servicing, and performance.

REQUIREMENT GUIDANCE

Air is cheap and universally available. Dry nitrogen may be preferred for stored-gas bottles because it is inert and generally available. For stored gas in missiles, helium may be preferred because of its compressibility factor which results in more energy at low temperatures (-65°F) than air or nitrogen. The selection of the gas medium should be made by the contractor, unless other factors dictate a unilateral choice by the procuring agency.

REQUIRMENT LESSONS LEARNED

Stored helium gas was selected for the Maverick missile. Analysis showed that helium is superior to air or nitrogen because of the better compressibility factor which results in greater energy at colder temperatures for a given stored gas volume. This fact was also demonstrated in 1980 on a development missile which used stored gas to deploy control surfaces. The control surfaces would not deploy at altitude with stored nitrogen gas, but did deploy when the nitrogen was replaced with helium.

Gas cylinders should be periodically analyzed for proper gas content. Flash fires occurred during door opening in a high-temperature oven containing a petroleum-fluid test system. Nitrogen-gas cylinders, used to provide an inert oven atmosphere, were analyzed and found to contain only 70 percent nitrogen.

4.2.1.2 Verification of gas medium. System gas medium selection shall be verified by analysis and inspection.

VERIFICATION RATIONALE (4.2.1.2)

Analysis should support the rationale for the gas medium selected. Criteria should include availability and supportability.

VERIFICATION GUIDANCE

The verification of gas selection shall be made by review of contractor's design and analysis report and drawings.

VERIFICATION LESSONS LEARNED

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3.2.1.3 System pressure. Main system operating pressure shall be _____ psi at _____ °F. Pressure transients during system operation shall not exceed _____ percent of operating pressure.

REQUIREMENT RATIONALE (3.2.1.3)

The system operating pressure should be designated at a specific temperature, especially for stored-gas bottle systems. Maximum pressures will dictate structural design criteria.

REQUIREMENT GUIDANCE

Unless otherwise defined, pressure transients exceeding 10 milliseconds should not exceed the proof pressure of the system. Pressure selection will affect availability of components, standardization, and testing parameters.

REQUIREMENT LESSONS LEARNED

4.2.1.3 Verification of system pressure. System pressure requirements shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.3)

Analyses should support the pressure selection. In case of suspected marginal performance, the analyses should be supplemented by tests at critical operating conditions.

VERIFICATION GUIDANCE

System pressure characteristics are generally verified by analysis of the contractor design report and drawings, laboratory tests of components, and system tests. When blowdown bottle systems are used, laboratory and flight tests are required to verify maintenance of adequate pressure to meet operating and/or life requirements.

VERIFICATION LESSONS LEARNED

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3.2.1.3.1 System pressure relief. Overpressure protection shall be provided.

REQUIREMENT RATIONALE (3.2.1.3.1)

Control of pneumatic system pressures is needed to prevent malfunctioning of pneumatic components and to prevent component structural failures if system pressures(s) become excessive.

REQUIREMENT GUIDANCE

If a pressure reducer is used to operate a subsystem at a reduced pressure, the subsystem should be capable of operating at the primary system pressure or have a subsystem pressure relief device to accommodate a pressure reducer malfunction.

The value could be a specific percentage of operating pressure, or taken from a table which lists pressure values (reference MIL-P-5518).

REQUIREMENT LESSONS LEARNED

4.2.1.3.1 Verification of system pressure relief. Pressure relief shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.3.1)

Pressure relief is necessary as a proof of safety.

VERIFICATION GUIDANCE

Pneumatic power system pressure relief control can be verified by analysis, inspection(s) and laboratory and system tests.

VERIFICATION LESSONS LEARNED

3.2.1.3.2 Back pressure. Proper functioning of any unit shall not be adversely affected by back pressure in the system.

REQUIREMENT RATIONALE (3.2.1.3.2)

Pneumatic power system back pressure must be limited to preclude performance degradation and system malfunction(s).

REQUIREMENT GUIDANCE

This is a concern in multiple subsystem pneumatic circuits.

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REQUIREMENT LESSONS LEARNED

To control damping back pressures in pneumatic system acutators using a 4-way valve control, the valve pressure poppets should open prior to return opening. Return gas is vented overboard. If the return is opened too early, damping back pressure is not available during the actuator stroke.

4.2.1.3.2 Verification of back pressure. Operating effects from back pressures shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.3.2)

Back pressure characteristics should be identified early during development to avoid redesign after full scale development.

VERIFICATION GUIDANCE

Proper operation with back pressures (if any) shall be verified by analysis, laboratory tests, and inspection of drawings.

VERIFICATION LESSONS LEARNED

3.2.1.3.3 System pressure indication. Indicator(s) shall be provided to measure pneumatic pressure.

REQUIREMENT RATIONALE (3.2.1.3.3)

Flight-station and/or maintenance personnel may need to know the operating status of pneumatic systems.

REQUIREMENT GUIDANCE

The pressure indicator(s) should be mounted in a position where they can be easily and accurately read.

REQUIREMENT LESSONS LEARNED

4.2.1.3.3 Verification of system pressure indication. Visibility and accuracy of system pressure shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.3.3)

The usefulness of pressure determination should be established.

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VERIFICATION GUIDANCE

The pressure indicating system should be verified by inspection and test. Visibility and accurate pressure indication should be verified by the ground and flight test programs.

VERIFICATION LESSONS LEARNED

3.2.1.4 Gas velocity limitations. Tubing size and gas velocity shall be selected to be consistent with design limits and to control actuator rates.

REQUIREMENT RATIONALE (3.2.1.4)

Actuation systems must operate at limited speeds and with sufficient deceleration characteristics to prevent structural damage. Although actuator speed is often controlled by restrictor or flow regulation and snubbing, line size can be a factor.

REQUIREMENT GUIDANCE

The system should operate within design limits which permit suitable control of actuator rates. For example, stopping or snubbing a cylinder piston at full stroke can be a design problem.

REQUIREMENT LESSONS LEARNED

The pneumatically powered B-29 bomb bay doors departed the aircraft during a ground demonstration because adequate snubbing was not provided to slow down the piston actuator after full stroke. This was corrected by restricting the air on the back side of the actuators to provide a snubbing air cushion.

The snubbing pressure of a pneumatic cylinder depends on the driving pressure and the time available for back-pressure bleed off. Trouble arises if the back pressure can bleed away before the damping function is complete. Use of the differential area for the effective output force ensures proper damping and maintains a controlled pressure in the damping section. Cylinders installed in such a system can be subjected to extension forces if high leakage rates occur across the piston seal. Canopy cylinder failures have been caused by air loads stalling the motion of the actuator during the closing cycle allowing the snubbing pressure to be lost. The original canopy cylinder design did not have differential area design. Canopies have also been jettisoned because of loss of damping air on the retract side of piston during the canopy opening cycle.

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4.2.1.4 Verification of gas velocity limitations. The operating characteristics dependent on gas velocities shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.4)

Operating characteristics peculiar to gas flow should be established.

VERIFICATION GUIDANCE

The gas velocity effects should be verified by analysis, inspection of drawings, and laboratory system tests.

VERIFICATION LESSSONS LEARNED

3.2.1.5 Pneumatic power system status indication. The operation of the pneumatic power systems shall be as automatic as practical; and the systems shall provide, but not be limited to, the following:

a. Information at the _____ which will alert the _____ to unsafe or marginal pneumatic power system operation so that appropriate corrective actions may be taken.

b. _____.

REQUIREMENT RATIONALE (3.2.1.5)

Flight-station or maintenance personnel need to know the status and/or operating condition of the pneumatic power system(s) prior to flight, and to be made aware of the status of the pneumatic power system during flight. The systems may be inoperative prior to actuation and this intelligence can provide alternate choices or corrective actions to the aircrew or maintenance personnel.

REQUIREMENT GUIDANCE

The blanks in (a) could be filled in with "flight station" and "flight crew" for an aircraft, or "test set" and "technician" for a missile. Other performance requirements, unique to a system, can be added as subparagraphs.

REQUIREMENT LESSONS LEARNED

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4.2.1.5 Verification of pneumatic power system status indication. Control and operating features of the pneumatic power system(s) shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.5)

The usefulness of power system operating status should be established.

VERIFICATION GUIDANCE

The provisions for system status indication should be verified by analysis, inspection and test.

VERIFICATION LESSONS LEARNED

3.2.1.6 Strength

3.2.1.6.1 Additional loads. During operation of the aircraft all pneumatic systems and components which are subjected to structural or other loads which are not of pneumatic system origin shall withstand the combination of loads that can be applied simultaneously in flight. There shall be no system or component failure or a permanent set, when the specified proof pressure and load combinations are applied.

REQUIREMENT RATIONALE (3.2.1.6.1)

The addition of external or other loads to pneumatic system pressure loads during normal operation may result in an overstress failure of the pneumatic power system.

REQUIREMENT GUIDANCE

The contribution of additive loads, such as structural deflections caused by flight maneuvers, should be identified. Examples are pneumatic system landing gear extension and/or landing loads.

REQUIREMENT LESSONS LEARNED

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4.2.1.6 Strength verification

4.2.1.6.1 Verification of additional loads. The ability to withstand additional loads shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.6.1)

The ability to withstand operating loads should be established.

VERIFICATION GUIDANCE

Additional loads should be evaluated by analysis and test.

VERIFICATION LESSONS LEARNED

3.2.1.6.2 Acceleration loads. Actuation cylinders, other components and their attaching lines and fittings, which in their functional operation are subjected to pressures generated by accelerated loads, shall withstand a pressure equal to 100 percent of the maximum pressure that will be developed without failure or permanent set at the maximum operating temperature.

REQUIREMENT RATIONALE (3.2.1.6.2)

Increased pressure caused by loads due to aircraft or component acceleration may cause premature component failure.

REQUIREMENT GUIDANCE

Components subject to acceleration loads should be identified to facilitate the design and application of adequate loads during qualification testing.

REQUIREMENT LESSONS LEARNED

4.2.1.6.2 Verification of acceleration loads. The ability of components to withstand pressures which are generated by acceleration loads shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.6.2)

The ability to withstand acceleration-induced loads should be established.

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VERIFICATION GUIDANCE

Testing is required to demonstrate that the pressure generated by acceleration loads is within allowable limits.

VERIFICATION LESSONS LEARNED

3.2.1.6.3 *Proof pressure.* No part of a system shall fail, take any permanent set, or be damaged in any manner, when subjected to applicable proof pressure of _____.

REQUIREMENT RATIONALE (3.2.1.6.3)

The intent of this paragraph is to assure that all systems and components are adequately designed and tested for functional integrity and structural integrity.

REQUIREMENT GUIDANCE

All system and components should be designed to withstand the proof pressure requirements of table II in MIL-P-5518, or other designated values, without permanent deformation and shall be fully functional following the test. When proof pressure tests are conducted at temperatures other than design temperatures, account for the degradation of material properties at design temperatures in determining the proof pressure.

It is recommended that proof loads be of sufficient magnitude to insure that durability requirements are met. (NOTE: The Air Force is currently evaluating the potential impact of this recommendation on weight, cost, performance, etc.). Periodic proof pressure tests at intervals less than one design service life may be considered subject to Air Force approval.

REQUIREMENT LESSONS LEARNED

4.2.1.6.3 Verification of proof pressure. Proof pressure compliance shall be verified as follows: _____.

VERIFICATION RATIONALE (4.2.1.6.3)

A material factor of safety should be established.

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VERIFICATION GUIDANCE

This requirement can only be demonstrated by actual tests on the system and components.

a. System. Proof pressure is applied to the system to prove the integrity of connections at each component. MIL-T-5522 may be used as a guide to perform this test.

b. Component. Proof pressure is applied to the component to prove the integrity of the component. MIL-P-8564 may be used as a guide for this test.

VERIFICATION LESSONS LEARNED

3.2.1.6.4 Burst pressure. No component shall rupture when subjected to applicable burst pressure of _____.

REQUIREMENT RATIONALE (3.2.1.6.4)

The intent of this paragraph is to ensure that all components are adequately designed and tested for structural integrity.

REQUIREMENT GUIDANCE

All pneumatic components for manned aircraft should be designed to withstand burst pressure requirements specified in table II of MIL-P-5518 without rupture. Deformation resulting from the test is acceptable, and the component need not be functional after the test. Components for missile applications may use lower burst factors if they are consistent with safety criteria (i.e. considering that they are subjected to less cumulative fatigue damage). When burst tests are conducted at temperatures other than design temperatures, account for the degradation of material properties at design temperatures in determining the burst pressure.

REQUIREMENT LESSONS LEARNED

4.2.1.6.4 Burst pressure verification. Burst pressure compliance shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.6.4)

The ability to withstand maximum design loads must be demonstrated.

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VERIFICATION GUIDANCE

Burst pressure is applied to the component to prove the integrity of the component. MIL-P-8564 may be used as a guide for the the test procedure. Hydrostatic pressure should be used for burst testing. Pneumatic pressure burst testing can be hazardous.

VERIFICATION LESSONS LEARNED

3.2.1.7 Environmental conditions. The pneumatic power system equipment shall be capable of withstanding or operating under the following conditions:

Environment	Requirement
Temperature	
Humidity	
Fungus	
Vibration	
Dust	
Salt Fog	
Explosive Atmosphere	
Acceleration	
Shock	

REQUIREMENT RATIONALE (3.2.1.7)

The intent of this requirement is to provide a suitable definition of environmental conditions from which the various pneumatic power system components must operate. Some components are sensitive to environmental considerations and some are not. With the environment defined, this judgment can be logically made on a component-by-component basis. If a certain environmental characteristic varies from one location on the airplane to another, the variation should be noted in this paragraph of the specification and the equipment evaluated accordingly.

REQUIREMENT GUIDANCE

Environmental requirements are contained in the various subsystem and component specifications. Each usually assumes the most adverse environment, and, much of the development efforts have been expanded trying to satisfy arbitrary conservative criteria. With a realistic combined environment defined, the development cycle can be tailored to meet the application. The risk is the ability to accurately assess the realistic combined environment. Consider the use of combined environmental reliability testing (CERT). Design equipment to known environmental stress loads with appropriate derating to assure adequate margins of safety to preclude failures.

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REQUIREMENT LESSONS LEARNED

4.2.1.7 Verification of environmental conditions. Environmental testing shall be conducted as follows: _____.

VERIFICATION RATIONALE (4.2.1.7)

The intent of test selection and test methods is to reflect the expected operating environment(s) for individual components and/or subsystems.

VERIFICATION GUIDANCE

Depending on the component, the environmental tests should be tailored to evaluate the critical characteristics of the component. The classical methods of environmental testing are presented in MIL-STD-810. These should be utilized whenever they can be determined to be applicable or near the operational duplication. The following standard methods are applicable:

Environmental	Requirement
Temperature	501, 502
Humidity	507
Fungus	508
Vibration	514
Dust	510
Salt Fog	509
Explosive Atmosphere	511
Acceleration	513
Shock	516

VERIFICATION LESSONS LEARNED

To get water collection characteristics, testing of pneumatic components for humidity and corrosion effects should include both static and operating conditions.

3.2.1.8 Electromagnetic interference. Electrical components shall meet the following requirements: _____.

REQUIREMENT RATIONALE (3.2.1.8)

Resistance to electromagnetic interference is referenced only for emphasis on pneumatic power system components since all electrical equipment on the aircraft must comply with uniform aircraft requirements.

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REQUIREMENT GUIDANCE

Requirements to ensure freedom from electromagnetic interference should be specified at the system level for use by all subsystems. MIL-STD-461 may be used as a guide.

REQUIREMENT LESSONS LEARNED

4.2.1.8 Verification of electromagnetic interference (EMI). Compliance with EMI resistance requirements shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.8)

The intent of these tests is to ensure acceptable performance in an electromagnetic environment.

VERIFICATION GUIDANCE

Component electromagnetic interference resistance should be verified by component tests in accordance with the system requirements. MIL-STD-462 and MIL-STD-826 may be used as guides.

VERIFICATION LESSONS LEARNED

3.2.1.9 Electrical bonding. Pneumatic power system tubing and components shall be electrically bonded in accordance with _____.

REQUIREMENT RATIONALE (3.2.1.9)

Bonding is essential in pneumatic power system design to minimize static charge accumulation and discharge.

REQUIREMENT GUIDANCE

The requirements of MIL-B-5087 or the Bonding and Lightning Protection Specification shall be applied.

REQUIREMENT LESSONS LEARNED

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4.2.1.9 Verification of electrical bonding. Electrical bonding requirements shall be verified by inspection and test.

VERIFICATION RATIONALE (4.2.1.9)

The intent is to ensure that desired conductive paths are available.

VERIFICATION GUIDANCE

Bonding of the pneumatic power system should be verified by compliance with the bonding specification(s).

VERIFICATION LESSONS LEARNED

3.2.1.10 Fire and explosion hazard proofing. Integration of the pneumatic power system in the vehicle shall preclude the possibility of creating a fire or explosion hazard.

REQUIREMENT RATIONALE (3.2.1.10)

Temperature and pressure conditions, if not controlled, could present a potential fire hazard.

REQUIREMENT GUIDANCE

Particular emphasis should be placed on the following: (a) pressure and temperature control, (b) holding of combustibles to a minimum, (c) if a compressor is used, avoiding excessively high compression ratios, and (d) if fire or explosion potential can't be avoided, providing flame arrestors or blow-out disks where possible.

REQUIREMENT LESSONS LEARNED

4.2.1.10 Verification of fire and explosion hazard proofing. Fire and explosion hazard proofing shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.10)

Analysis and inspection of drawings should show protection features. Tests may be required in some instances.

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VERIFICATION GUIDANCE

The Fire and Explosion Hazard Protection specification shall be used for guidance.

VERIFICATION LESSONS LEARNED

3.2.1.11 Survivability - vulnerability (S-V). The pneumatic power systems shall be configured to accept a single impact by the threat(s) specified in the System or Air Vehicle Specification without loss of defined mission-essential functions.

REQUIREMENT RATIONALE (3.2.1.11)

Aircraft performance limitations after pneumatic power system(s) damage should be defined.

REQUIREMENT GUIDANCE

Capability should be defined for loss of systems, as well as impact by specific threat(s). Refer to the Mission Completion in a Combat Environment Specification for guidance. Failure of a backup emergency pneumatic system should not prevent operation by the normal (hydraulic or electric) system. However, for most functions, capability to operate after failure of both the normal and emergency power systems is not required.

REQUIREMENT LESSONS LEARNED

4.2.1.11 Verification of survivability - vulnerability (S-V). S-V of the pneumatic power systems shall be verified as follows: _____.

VERIFICATION RATIONALE (4.2.1.11)

S-V requirements will be verified per the analysis and test requirements of the system specification.

VERIFICATION GUIDANCE

Survivability - vulnerability characteristics should be verified by analysis and test.

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VERIFICATION LESSONS LEARNED

3.2.1.11.1 Resistance to gunfire. Pneumatic power system pressure vessels shall not result in fragmentation from a single impact by _____.

REQUIREMENT RATIONALE (3.2.1.11.1)

Gunfire damage to a pressure vessel should not result in fragmentation which can injure personnel or cause serious damage to the airframe.

REQUIREMENT GUIDANCE

High-pressure vessels have been required to take a single armor-piercing incendiary (API) .50 caliber projectile without fragmenting.

Low-pressure vessels may require resistance to a single .30 caliber API projectile.

REQUIREMENT LESSONS LEARNED

Experience has shown that AISI 4130 and comparable steels should not be heat-treated above 170,000 psi to preclude fragmentation.

4.2.1.11.1 Verification of resistance to gunfire. Pressure vessel resistance to fragmentation shall be verified as follows: _____.

VERIFICATION RATIONALE (4.2.1.11.1)

Actual tests are required to show resistance to gunfire.

VERIFICATION GUIDANCE

Resistance to fragmentation should be evaluated by test. Past practice (reference MIL-P-8573) has been: For reservoirs three (3) inches or less in diameter, the ammunition should be .30 caliber M-2 armor piercing. For all other sizes, the ammunition should be .50 caliber M-2 armor piercing. Muzzle velocity should be 2800 fps and range 50 yards maximum. The reservoir should be positioned with its longitudinal axis 45° from normal away from the gun position. The projectile should be tumbled so that the .30 caliber produces an "in" hole at least 1/4 inch wide by 5/8 inch long and the .50 caliber should produce an "in" hole at least 1/2 inch wide by 1 1/2 inch long. The projectile velocity after tumbling should be monitored during the last 10 feet prior to impact. Minimum velocity should be 2300 fps. The reservoir should remain in one piece, and the material of the cylinder should not tear excessively in any one direction.

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VERIFICATION LESSONS LEARNED

3.2.1.11.2 Component clearance. Clearance shall be maintained between moving system components and structure or other components to ensure that no possible combinations of temperature effects, airloads, wear, or structural deflections can cause binding, rubbing, or jamming. There shall also be adequate clearance for component installation, removal, and maintenance.

REQUIREMENT RATIONALE (3.2.1.11.2)

The intent of this requirement is to prevent binding, chafing, or jamming of components in static or operating modes. In addition, there are minimum clearances necessary to accommodate removal and installation of components.

REQUIREMENT GUIDANCE

Component installation should have separation clearances which are maintained under all aircraft operating modes. There should be clearance for hand tools required to remove and install components. There should be sufficient space to remove and install components. This may require additional access panels and doors. Refer to SAE ARP 994 for detail guidance.

REQUIREMENT LESSONS LEARNED

4.2.1.11.2 Verification of component clearance. Clearance(s) requirements shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.11.2)

The best way to determine clearance is by close examination of installation mockup(s), and by examination of the air vehicle, before and after flight tests. Tests may by required to demonstrate maintenance of adequate clearances during dynamic operation, e.g., landing gear, doors, etc. and structural deflections caused by air loads, landing loads, etc.

VERIFICATION GUIDANCE

Separation clearance of components under all static or operating modes should be verified by analysis, inspection, and test.

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VERIFICATION LESSONS LEARNED

3.2.2 Functional components characteristics

3.2.2.1 Components selection. Components of the pneumatic power system shall be _____.

(a. Standard components. Military standard components shall be used whenever they satisfy the system operating needs.)

REQUIREMENT RATIONALE (3.2.2.1a)

The use of standard components is in agreement with Air Force policy to minimize supply problems and reduce development, procurement, and logistics costs. The use of proven components should result in greater reliability.

REQUIREMENT GUIDANCE

The use of standard equipment should be emphasized and encouraged. If there are special procedures for selection of standards and specifications, this should be included.

REQUIREMENT LESSONS LEARNED

(b. Nonstandard components. Nonstandard component design and performance requirements shall be defined in the contractor's specifications.)

REQUIREMENT RATIONALE (3.2.2.1b)

A logical process is necessary to define the performance requirements for nonstandard components.

REQUIREMENT GUIDANCE

Nonstandard components should be designed and tested in accordance with the contractor's specifications. The contractor should negotiate the number of cycles of operation for nonstandard components with the procuring activity. Ground and flight operating cycles should be included. If a suitable military standard component is not available, contractor designed equipment which has been approved and/or proven in an aircraft pneumatic power system application should be considered instead of developing new components. Critical nonstandard and component specifications are normally submitted to the procuring activity for approval.

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REQUIREMENT LESSONS LEARNED

4.2.2.1 Verification of component selection. Component selection shall be verified by _____.

(a. Verification of standard components. Use of standard components shall be verified by _____.)

VERIFICATION RATIONALE (4.2.2.1a)

This requirement can be accomplished by reviews of the system schematic and related parts lists.

VERIFICATION GUIDANCE

This requirement can be verified in several ways, depending on the scope of the program, e.g. parts review board. This requirement is generally verified by inspection.

VERIFICATION LESSONS LEARNED

(b. Verification of nonstandard components. The verification of nonstandard component(s) performance shall be by satisfactory compliance with the contractor's component specifications.)

VERIFICATION RATIONALE (4.2.2.1b)

This can be accomplished by review of contractor's drawings and evaluation of the contractor's test report(s) and acceptance test plans.

VERIFICATION GUIDANCE

Nonstandard components verification should be identified to ensure compliance with total system performance requirements.

VERIFICATION LESSONS LEARNED

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3.2.2.2 Actuators. Actuator design and performance requirements shall be defined in the contractor's specification(s). Provision shall be made for control of rates of operation for resisting, assisting, and no load conditions.

REQUIREMENT RATIONALE (3.2.2.2)

Actuators are usually nonstandard components with unique requirements.

REQUIREMENT GUIDANCE

The actuator performance, design, and verification requirements should be established in accordance with MIL-STD-490.

The problem areas with linear actuators have been:

a. Fatigue, especially with aluminum parts

b. Rod end fitting backout

c. Structural deformation of attach point; also bearing distortion

d. Rod end attachment cracking at thread relief

e. Rod seal leakage

f. Bore wear in aluminum barrels

g. Leakage through "faulty" chrome plating.

h. Column buckling of piston rods.

General considerations in actuator design module:

a. Use of steel barrels

b. Use of generous radii (to avoid stress risers)

c. Use of rolled threads when practical

d. Provision of positive attachment and antirotation locking of the rod end. Ordinary jam locknuts have not been sufficient.

REQUIREMENT LESSONS LEARNED

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4.2.2.2 Verification of actuators. Actuator performance shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.2)

Tests which represent system function and operating loads are required.

VERIFICATION GUIDANCE

Verification of actuator design and performance should be based on successful compliance with the contractor's component verification program.

VERIFICATION LESSONS LEARNED

3.2.2.3 Multiple control valves. In systems which incorporate two or more directional control valves, provision shall be made to prevent the medium from being transferred inadvertently, at any possible valve setting, from the cylinder ports of one valve into the cylinder ports of another valve.

REQUIREMENT RATIONALE (3.2.2.3)

The control of the air flow is required so that system operation will not be affected by interflow between valves.

REQUIREMENT GUIDANCE

Directional control valve(s) installation and operation(s) should be analyzed.

REQUIREMENT LESSONS LEARNED

4.2.2.3 Verification of multiple control valve(s) operation. The integrity of multiple control valve operation shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.3)

This can be evaluated by analysis and functional system test(s).

VERIFICATION GUIDANCE

Multiple control valve use should be based on approval of the contractor's specification(s), system analysis, design report, and tests.

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VERIFICATION LESSONS LEARNED

3.2.2.4 Filters. There shall be means to filter all pneumatic power system gas during flight, ground, and servicing operations. Filter elements shall be easily replaced and shall be of the disposable type.

REQUIREMENT RATIONALE (3.2.2.4)

Filters are required to preclude excessive wear and malfunction of pneumatic components such as motors, directional control valves, etc. It is essential for performance and life that most particulate particles be removed from the gas.

REQUIREMENT GUIDANCE

At least 25-micron absolute filters should be provided in all systems. These filters should be used to filter all the gas supply in the system. Reservoir filters should be of the replaceable element type in refillable or rechargeable systems. The total gas supply should be filtered through a 25-micron absolute filter before use in the system. In some system(s) finer filtration, less than 25-micron absolute may be required. Experience has shown that maintenance personnel must be continuously aware of the importance of keeping contamination out of pneumatic power systems. The maintenance technical orders should provide adequate instructions to prevent system contamination when the systems are opened for maintenance and inspections.

REQUIREMENT LESSONS LEARNED

4.2.2.4 Verification of filters. The required use and effectiveness of filters shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.4)

This is accomplished by system design review, evaluation of filter qualification compliance, and service experience.

VERIFICATION GUIDANCE

The filter effectiveness can be verified in conjunction with the aircraft ground and flight test programs.

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VERIFICATION LESSONS LEARNED

3.2.2.5 Moisture content control. Moisture content shall be limited to _____.

REQUIREMENT RATIONALE (3.2.2.5)

The moisture content in the gas should be held to a low value to preclude corrosion-induced failures, or ice formation.

REQUIREMENT GUIDANCE

Past practice has been to limit air discharged from dehydrators used in airborne-compressor-charged systems to no more than 0.12 grains of moisture per pound of dry air (specific humidity).

Drains should be placed at low points in the system to periodically drain accumulated water.

Do not install components in low point(s) in system. Moisture froze in a solenoid valve which was installed at a low point in a bomb bay door system. The solenoid valve would not operate.

Chemical driers should be large enough to meet specifications for moisture removal. They should not require too frequent cartridge removal under the most adverse conditions. Use of the pneumatic power system, such as for normal or emergency functions, is a determining factor. Drier housing and cartridge housing materials should resist corrosion (Example: Plastic shell cartridge). Dissimilar metals should be avoided because of electrolytic action. Use built-in filters to prevent desiccant powder migration. The unit should be rigidly mounted to facilitate servicing. Although molecular-sieve is frequently used in aircraft over silica-gel as a desiccant because of its greater moisture holding ability, it can attack aluminum in the presence of water. Therefore, maximum corrosion protection is required. Visual moisture indicators would simplify cartridge removal time and save on cartridges.

Chemical drier problems:

a. Cracked housing. Fix: material change.

b. Corrosion throughout unit. Fix: plastic cartridge. Also, consider sealing cartridge in housing at inlet end instead of outlet end. This will allow cartridge housing to be pressurized with dry air to avoid corrosion.

c. Desiccant particle migration. Fix: better cartridge filter. Porous bronze filter allowed desiccant migration.

d. Rigidity of mounting inadequate for servicing. Fix: Rigid mount capable of withstanding wrenching torques.

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REQUIREMENT LESSONS LEARNED

4.2.2.5 Verification of moisture content control. Moisture removal effectiveness shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.5)

This can be determined by measuring actual moisture content in a representative air sample.

VERIFICATION GUIDANCE

Moisture removal effectiveness can be verified in conjunction with component testing, and aircraft ground and flight test programs.

VERIFICATION LESSONS LEARNED

3.2.2.6 Fittings. Straight-threaded fittings shall be used. Unless specifically approved by the procuring activity, no thread lubricants shall be used on straight-threaded fittings. All nonstandard fittings shall require approval by the procuring activity.

REQUIREMENT RATIONALE (3.2.2.6)

Standardization of fitting threads is essential. MIL-S-8879 threads are preferred for high-stress applications but are more costly to produce. Fitting threads in accordance with MIL-S-7742 have proven to be satisfactory for many years. Military standard fittings should be used to reduce the inventory of nonstandard fittings.

REQUIREMENT GUIDANCE

Only straight threads shall be used for reconnectable fittings. The use of permanent-type fitting connectors employing no screw threads requires procuring activity approval. In the late 1940s, the 37-degree cone "flared" fitting was the only tube fitting standard and required mechanical flaring of the connecting tube. In the 1950s, the flareless fitting was made available as an alternate standard because of its increased resistance to vibration. In the 1960s, the introduction of higher strength tubing made the standard flared and flareless fitting inadequate, because the harder tubing is difficult to flare, and also too hard for the flareless sleeve to penetrate the tube surface. New fittings were introduced for attachment to the harder tubing. This has created changes such as special tools and mechanical equipment to assemble the fittings to the tubing. The procuring activity should either specify tube fittings or consider a trade study.

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REQUIREMENT LESSONS LEARNED

4.2.2.6 Verification of fittings. Fitting selection shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.6)

Although weight reduction and reliability are prime requirements, there are other considerations in life cycle costing.

VERIFICATION GUIDANCE

Selection of fittings should be based on evaluation of trade studies, if conducted, and inspection.

VERIFICATION LESSONS LEARNED

Pipe-thread fittings tend to leak after repeated assembly and disassembly and were discontinued for use in aircraft. Permanently installed pipe-thread fittings were permitted as plugs, but today there are other accepted methods for aircraft component plugs. Use of pipe-threaded plugs for external sealing of drilled passages can set up internal stresses in component houses resulting in cracks.

Flared fitting sleeves of aluminum bronze were subject to stress corrosion cracking and were prohibited for use.

Aluminum flared fittings should not be used with steel tubing in sizes smaller than -8 because of the low relative yield strength of aluminum.

Aluminum fittings and aluminum tubing are not permitted in designated fire zones.

The use of free-machining steels such as AISI 1137 and 1141 have caused problems because of marginal yield strength when subjected to repeated fitting assembly.

3.2.2.7 Packing installation. _____ shall be used as guides for packing installation. Grease conforming to MIL-G-4343 shall be used as a lubricant for Buna N type packings in pneumatic power systems.

REQUIREMENT RATIONALE (3.2.2.7)

Good gland design controls the "O" ring squeeze that is required to obtain leak-free packing installations and adequate packing life over the temperature range of the system. Grease conforming to MIL-G-4343 is used to reduce packing friction and to provide some plasticization of the Buna N packing material.

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REQUIREMENT GUIDANCE

A guide for gland design and packing installation is MIL-G-5514.

REQUIREMENT LESSONS LEARNED

4.2.2.7 Verification of packing installation. Packing installation design shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.7)

A review of detail dimensions for the packing and packing cavity is a preferred method for seal installation design analysis.

VERIFICATION GUIDANCE

The packing installation effectiveness can be evaluated by inspection and appropriate tests.

VERIFICATION LESSONS LEARNED

3.2.2.8 Seals. Packings and gasket materials shall be compatible with the fluid medium and capable of performing under the specified environment.

REQUIREMENT RATIONALE (3.2.2.8)

Packing and gasket materials must have good compatibility with the fluid medium so that their sealing performance and life are not compromised in the specified environment.

REQUIREMENT GUIDANCE

MIL-G-5510 and MIL-P-83461 may be used as guides for materials compatible with the fluid medium. MIL-P-8564 may be used for guidance for pneumatic component design.

REQUIREMENT LESSONS LEARNED

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4.2.2.8 Verification of seals. Seal compatibility and compliance with leakage requirements shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.8)

The AFWAL Materials Laboratory is one source for data on fluid-elastomer seal compatibility. Allowable leakage is a function of system needs and ground servicing criteria.

VERIFICATION GUIDANCE

Packing and gasket performance can be verified by the component qualification tests and the aircraft ground and flight test programs.

VERIFICATION LESSONS LEARNED

3.2.2.9 Tubing systems

3.2.2.9.1 Tubing system durability. Tubing and tube connectors shall have a system durability life equal to or greater than the life of the aircraft.

REQUIREMENT RATIONALE (3.2.2.9.1)

Because of the cost and downtime required for tubing replacement, tubing and tubing connectors should have a system life equivalent to the life of the aircraft.

REQUIREMENT GUIDANCE

When new tubing material or tube fittings are proposed, endurance-limit tests should be conducted. It is also recommended that SAE ARP 994 be used for guidance.

REQUIREMENT LESSONS LEARNED

4.2.2.9 Verification of tubing systems

4.2.2.9.1 Verification of tubing systems durability. The durability of tubing systems shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.9.1)

Durability evaluation includes tubing with nicks, scratches, and out-of-round cross-section(s) caused by tube bending operations and installation.

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VERIFICATION GUIDANCE

Tubing and tubing connector durability should be evaluated by analysis, inspection, and test. Parameters to be considered are permanent vs threaded connectors, repeated assembly characteristics, envelope for installation, torque characteristics, materials, tube dimensions (wall thickness), tube support, and operating environments.

VERIFICATION LESSONS LEARNED

Early tubing failures have occurred in areas of extreme vibration. Engine nacelle and pylon areas can induce resonance which can fail tube supports and/or tubing. For example, in the engine area of one type aircraft, tubing failures occurred within 90 flight hours. The very early failures were caused by a combination of resonant frequency, marginal strength loop clamps, and tubing cross section which had excessive ovality (out-of-roundness at the tube bend). Corrective action was to change from carbon-steel clamps to stainless-steel clamps, add additional loop clamps, use some block clamps, and revise quality assurance procedures.

3.2.2.9.2 Tubing identification. All pneumatic lines shall be identified by _____.

REQUIREMENT RATIONALE (3.2.2.9.2)

For maintenance purposes, the pneumatic lines should be identified (1) as pneumatic lines and (2) for function e.g. landing gear down, flaps down, etc. Identification is required as often as necessary, to track the line function, particularly on lines entering and emerging from closed compartments.

REQUIREMENT GUIDANCE

Pneumatic lines should be identified in accordance with MIL-STD-1247. Other unique requirements can be added.

REQUIREMENT LESSONS LEARNED

4.2.2.9.2 Verification of tubing identification. Tubing identification shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.9.2)

The best method of verification is inspection of tubing runs in the aircraft.

VERIFICATION GUIDANCE

Pneumatic line identification should be verified by inspection.

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VERIFICATION LESSONS LEARNED

3.2.2.9.3 Tubing clearance. Clearance shall be maintained to prevent physical contact of tubing with other tubing, components, or structure as a result of system operation, flight maneuvers, or test pressures. There shall be adequate installation clearance and support for wrenching and removal.

REQUIREMENT RATIONALE (3.2.2.9.3)

The intent of this requirement is to prevent contact, rubbing, or abrasion of installed tubing in static and operating conditions. In addition, there are minimum clearance dimensions necessary to accommodate wrenches and other tools required for connecting mating tubes.

REQUIREMENT GUIDANCE

Minimum clearances should take into consideration: vibration, wear, structural deflection, manufacturing tolerances, and tube movement due to temperature changes or pressure surges. For guidance, refer to SAE ARP 994, Paragraph 5.2, Clearances.

REQUIREMENT LESSONS LEARNED

There have been many system failures and aircraft mishaps caused by rubbing contact of line(s). Examples are:

a. Maintenance T.O. did not include instructions for the required orientation of a rigid tube assembly replacement.

b. Chafing of electrical wiring causing arcing and burn through of the rigid tubing.

c. Structural deflection of bulkhead caused by pressurization or vibration.

d. Titanium tubing was worn through under a tube support loop clamp in a high vibration environment.

4.2.2.9.3 Verification of tubing clearance. Tubing clearance shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.9.3)

The best way to determine clearances is by examination of installation mockup, and by examination of air vehicle installations, before and after flight tests. Tests may be required to demonstrate maintenance of adequate clearances during operation, e.g. landing gear, doors, etc. and structural deflection caused by air loads, landing loads, etc.

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VERIFICATION GUIDANCE

It is recommended that clearances be verified by inspection, analysis, and tests.

VERIFICATION LESSONS LEARNED

3.2.2.9.4 Repair and replacement of tubing. Repair and replacement methods for tubing and fittings shall be established. This information shall be included in appropriate technical manuals and/or technical orders.

REQUIREMENT RATIONALE (3.2.2.9.4)

A recommended procedure is required for repair and replacement of damaged sections of pneumatic lines by maintenance personnel at organizational levels.

REQUIREMENT GUIDANCE

Repair methods shall be established for section damage of rigid lines and included in technical manuals and/or appropriate technical orders for the particular aircraft.

REQUIREMENT LESSONS LEARNED

It has been a standard practice, when a factory-made replacement is not available, to make full tubing replacement with 304 1/8 Hard tubing which is easier to form than the high-strength AM350 or 21-6-9 tubing material. Partial tube replacement has been with 304 1/8 Hard tubing and Permaswage fittings. Titanium tubing repair may require special procedures.

4.2.2.9.4 Verification of repair and replacement of tubing. Methods for repair and replacement of tubing and fittings shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.9.4)

Tubing repair and replacement methods must be consistent with Air Force maintenance practices.

VERIFICATION GUIDANCE

Repair and replacement methods should be verified by demonstration, inspection, and technical order validation.

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VERIFICATION LESSONS LEARNED

3.2.2.10 Relative motion. Means shall be provided to accommodate relative motion.

REQUIREMENT RATIONALE (3.2.2.10)

Components or other installations that have relative motion require flexible connections to permit freedom of operation.

REQUIREMENT GUIDANCE

Relative motion between two points is usually provided by:

a. Hose assemblies: Standard TFE (teflon) hose assemblies are preferred to rubber hose assemblies, especially when age limitations and high temperatures are a factor. Proper routing of hoses is critical. Allow for flexibility without binding or crimping. Don't use hose(s) under constant pneumatic pressure because of gas permeability with time. Use SAE ARP 994 as a guide.

b. Coiled tubing: Use SAE ARP 584 as a guide.

c. Swivel joints: Use MIL-J-5513 as a guide. The requirement may be modified to restrict the type of flexible connections, if desired.

REQUIREMENT LESSONS LEARNED

Hose assemblies:

Rubber hose assemblies have leaked or failed because of heat aging. Teflon hose should be used in higher temperature applications. If rubber hose is used, it should not exceed the recommended temperature environment, and should be inspected on a periodic basis. Hose assemblies, like rigid tubing should be installed for chafing resistance by making allowance for minimum clearance during normal operation and structural deflection and/or vibration.

Chafe-resistant sleeves are available but should not be used as a substitute for good installation practices.

Coiled tubing: Coiled tubing has been used successfully in commercial and military aircraft installations. Coiled tubing has been used most successfully in tube diameters of 3/8 inch or less.

Swivel joints: Swivel joint leakage has been a problem. Nonplanar actuation (side loading on the swivel joint) is a contributing factor. C-141 swivel joints were susceptible to dirt entering the seal cavity. Laboratory tests were conducted on the swivel joints with dust excluders and the leakage problem was significantly reduced.

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4.2.2.10 Verification of relative motion. Means to provide relative motion shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.10)

Relative motion must be accomplished without stress type failures or chafing.

VERIFICATION GUIDANCE

Provisions for relative motion should be verified by inspection, demonstration, and test.

VERIFICATION LESSONS LEARNED

3.2.2.11 Air compressor(s). When an airborne compressor package(s) is used, general and specific requirements shall be defined in a Critical Item Development specification.

REQUIREMENT RATIONALE (3.2.2.11)

The operating characteristics and associated equipment for a compressor package must be defined in detail.

REQUIREMENT GUIDANCE

Specification MIL-C-6591, "Compressor Package, Air, Aircraft, Electric-Motor-Driven, General Specification For" can be used for guidance for either an electric-motor-driven package, or for a compressor (engine-driven or hydraulic-motor-driven). A compressor package may include:

a. Driven air compressor

b. Relief valve

c. Moisture separator

d. Chemical dehydrator

e. Check valve

f. Priority (or back-pressure valve)

g. Means for periodic draining of the moisture separator

h. Pressure switch and relay

i. Cooling means to prevent exceeding maximum operating temperature.

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Experience has led to the following criteria:

a. The moisture separator shall provide conditioned air to the reservoir which is relatively moisture-free to prevent blocking of expansion lines by freezing. At rated discharge pressure, and air entering the inlet at sea level pressure, and a dewpoint temperature of +85°F, the dewpoint of the discharge air shall not exceed a specified maximum value based on sea level pressure.

b. Automatically, at intervals specified in the air compressor package specification and at shutdown, the condensate in the moisture separator shall be purged to atmosphere. During the purging process, there shall be no backflow of air from down stream of the moisture separator. The moisture separator shall be provided with a heater that operates concurrently with the aircraft operation to prevent any remaining condensate from freezing, and shall be thermostatically controlled.

c. One or more blow-out disks shall be incorporated as necessary to provide against possible explosions within the compressor package.

d. Pressure switch contacts shall be snap action and shall be moisture sealed.

e. The compressor package shall be lubricated from its own oil supply.

REQUIREMENT LESSONS LEARNED

At altitude, there may be insufficient inlet air for compressor operation.

Supercharging of the compressor inlet may be necessary for adequate performance.

Another problem at altitude, or even at sea level, is adequate ram-air cooling flow. At one time, the F-84 package was located in the aircraft wing root where ram-air cooling flow was insufficient.

4.2.2.11 Verification of air compressor(s). Verification of the air compressor package shall be by inspection, demonstration and tests.

VERIFICATION RATIONALE (4.2.2.11)

The integrity and operating characteristics of the compressor package must be identified.

VERIFICATION GUIDANCE

Tests should include conformance to the requirements of the approved compressor package specification and appropriate ground and flight tests. Special tests or criteria can be added if desired.

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VERIFICATION LESSONS LEARNED

3.2.2.12 Air bottles. When air bottles are used, the general and specific requirements, quality assurance provisions, and preparation for delivery requirements shall be defined in a Critical Item Development specification.

REQUIREMENT RATIONALE (3.2.2.12)

Air bottles are safety-critical items.

REQUIREMENT GUIDANCE

Use MIL-P-5518 and MIL-R-8573 for guidance. Unique characteristics can be added when appropriate.

REQUIREMENT LESSONS LEARNED

Gas bottles should be located in an area where bottle failure will not result in personnel exposure to fragmentation and will minimize structural damage (see 4.2.1.11.1, Resistance to Gunfire). Materials impact:

a. A SRAM missile helium bottle ruptured in a stored missile. Failure was attributed to stress corrosion which may have been caused by cleaning agent(s) residue. Room-temperature pressure was 7,700 psi; burst pressure, 24,000 psi, material was 18 Ni 200 maraging steel.

b. F-102, F-106 filament-wound plastic pressure bottle explosions occurred at 2,100 psi during filling on the ground. Maximum allowable pressure was 3,100 psi. The bottle(s) increased considerably in circumference when pressurized. Loss of strength occurred after repeated proof pressure tests (circa 1959).

c. A 5,000-psi nitrogen storage bottle, of fiberglass S-Glass/epoxy composite structure, ruptured and caused major aircraft damage. The bottle had been pressurized for more than eight years. Failure was attributed to the phenomenon known as "stress rupture." Stress rupture of S-glass has been extensively investigated by Dr. T. T. Chiao and associates of the Lawrence Livermore National Laboratory. Emperical data shows an inverse relationship between the safe life of S-glass/epoxy composite structures and the ratio of loading stress to ultimate stress of the composite structure. A discussion of the stess rupture phenomenon in composites is presented in a technical article by T. T. Chiao and R. L. Moore entitled "Stress-Rupture of S-Glass/Epoxy Multi-Filament Strands" published in the Journal of Composite Materials, Vol. 5, 1971. G. Shaughnessy and J. A. Snide of the University of Dayton have prepared a report entitled, "Statistical Treatment of S-Glass/Epoxy Stress Rupture Data." This report contains tabular life data as a function of the fiber stress, quantile level, and confidence interval. It is available from the Materials Laboratory of the Air Force, Wright Aeronautical Laboratories (AFWAL/MLSE, Wright Patterson AFB OH 45433).

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Consider the following points in air bottle design and installation:

a. Use AN6024 type moisture bleed valves. AN814 fitting results in "O" ring blow out during bleeding.

b. Proper bottle positioning and aircraft access for water purging are required.

c. Avoid allowing water to spray on electronic equipment.

d. Installations requiring moisture pick-up tubes should have a flex tube long enough to reach the lowest point of bottle. Use a plastic end to avoid bottle interior coating damage.

e. Don't drop sharp objects on fiberglass bottles.

f. Problems have occurred with the internal coating inside air bottles flaking off and then obstructing the control valve. Studies showed the problem was caused by rapid ground charging of the air bottles. The bottles were being charged on the ground with an external source in approximately one second. The temperature inside the bottle would be over 400°F which would cause the internal coating to flake off. The solution was to charge the bottles more slowly (approximately 1 minute for 0 to 3000 psi).

4.2.2.12 Verification of air bottles. Air bottle design and performance shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.12)

High pressure gas storage bottles have a hazard potential if not properly designed and utilized.

VERIFICATION GUIDANCE

Verification of air bottle design and performance should be based on inspection, demonstration, and compliance with the contractor's Air-Force-approved test program(s). Other unique test criteria can be added in the blank as appropriate. Ground and flight tests should be a major part of component and system validation.

VERIFICATION LESSONS LEARNED

3.2.2.13 Electric-motor-driven compressor. Electric-motor-driven equipment shall be capable of continuous operation.

REQUIREMENT RATIONALE (3.2.2.13)

Experience has shown that continuous-duty electric motors are required.

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REQUIREMENT GUIDANCE

Other requirements for electric-motor-driven equipment such as test duty cycle(s), operational temperature range, rated voltage, altitude, explosion proofing and, automatic control vs manual control should be defined. MIL-M-7969 and MIL-M-8609 can be used as a guide for preparation of electric motor design requirements.

REQUIREMENT LESSONS LEARNED

Electric motor-driven equipment has been subject to early failures when the electric motors were rated for intermittent duty. For reliability and long life, it is necessary to have electric motors with continuous-duty capability.

4.2.2.13 Verification of electric-motor-driven compressor. Performance of the electric-motor-driven equipment shall be verified by component and sub-system tests.

VERIFICATION RATIONALE (4.2.2.13)

The integrity and durability of the electric-motor system is critical to compressor operation.

VERIFICATION GUIDANCE

The procurement specification(s) should define the inspection and tests required to verify compliance with the specified performance and durability requirements.

VERIFICATION LESSONS LEARNED

3.2.2.14 Shelf life. All pneumatic power system components, when properly prepared for storage, shall have a minimum shelf life of _____.

REQUIREMENT RATIONALE (3.2.2.14)

The shelf life of components must be controlled so that performance and operational life will not be compromised.

REQUIREMENT GUIDANCE

Elastomeric items should, when properly prepared for storage, have a minimum shelf life in accordance with MIL-STD-1523.

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REQUIREMENT LESSONS LEARNED

4.2.2.14 Verification of shelf life. Component shelf life capability shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.14)

Identification of shelf life items is required for maintenance planning and spare parts provisioning.

VERIFICATION GUIDANCE

Shelf life should be evaluated by analysis, inspection, and service experience.

VERIFICATION LESSONS LEARNED

3.2.2.15 Endurance. The components of the pneumatic power system shall have a total minimum life equal to _____, except for: _____. Endurance cycling life of components shall be established.

REQUIREMENT RATIONALE (3.2.2.15)

Mechanical elements of the pneumatic power system should have a minimum total useful life equal to the useful life of the overall aircraft with state-of-the-art servicing.

REQUIREMENT GUIDANCE

Some individual components may be overhauled or replaced at intervals shorter than the life of the overall aircraft. This may include parts subject to wear such as seals, and bearings and parts subject to replacement such as moisture-removal agents and filter elements. For structural elements, the safe operational life and required maintenance intervals should be developed using damage-tolerance approaches similar to those developed for new airframe structures and successfully applied to older in-service aircraft (i.e., considering fracture-mechanics principles) as described in MIL-STD-1530, MIL-A-83444 and the General Specification For Airframe Structures. This approach considers the potential for initial flaws occurring during manufacture and/or maintenance and consists of the determination of the anticipated usage spectrum and the potential growth of these initial flaws to failure or until detected prior to failure by normal in-service inspections.

Each component in the system (e.g., actuators, reservoir, tubing, etc.) should be examined for consequence of failure or hazard, and safety of flight.

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It is assumed that each component is subjected to a proof test per 3.2.1.6.3 It is recommended that pressure vessels be designed to leak before they fail (i.e., the critical crack size at maximum operating pressure is greater than the critical element dimension in a direction of a leak path). For this situation, leakage should be immediately obvious to ground personnel conducting routine maintenance and/or detectable with low-pressure warning devices installed.

Initial flaws should be developed on the basis of manufacturing nondestructive inspection (NDI) or proof testing as appropriate. (Note: see section on proof test requirements.) When proof tests are conducted, the initial flaw assumption shall be a worst-case shape, severity (i.e., a long, shallow surface flaw) and location considering the test ambient temperature and the upper-bound fracture toughness.

It is recommended that elements be designed so that initial flaws will not grow to failure at maximum service loading within the design service life of the airframe considering anticipated service loads, functional loads, proof loads, and vibratory loads anticipated throughout this period, and without the requirement for in-service inspection (other than that associated with leak before fail).

It is recommended that all elements (including those that meet leak before break criteria) be designed to at least one design lifetime.

For safety-of-flight components or components with hazard potential (without a leak-before-fail characteristic), it is recommended that the design life be two times the service life of the airframe. For noncritical components, it is recommended that the design life be equal to the service life of the airframe. For leak-before-break situations, it is recommended that the design life before leakage be at least one design service life of the aircraft. It is recommended that components be designed to preclude sustained load-crack growth considering flaw sizes at the end of one lifetime and the thermal and chemical environment to which the component is subjected in service.

REQUIREMENT LESSONS LEARNED

4.2.2.15 Verification of endurance. System and component operating life shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.15)

Durability criteria is required for vehicle service life planning and/or periodic in-service inspections.

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VERIFICATION GUIDANCE

The individual component specification should define the testing required to verify its useful life.

VERIFICATION LESSONS LEARNED

3.3 Reliability. The pneumatic power system reliability requirements and allocations shall be derived from the overall system reliability program.

REQUIREMENT RATIONALE (3.3)

The pneumatic power system reliability is an integral part of the reliability requirements for the total air vehicle. Therefore, proper allocation of reliability with the associated confidence level should be generated for the system. This then must be further allocated to the major components or subsystems as described by the basic air vehicle specification.

REQUIREMENT GUIDANCE

The overall system reliability monitor should assist in establishing these requirements. For details relating to reliability programs, consult MIL-STD-785.

Component or subsystem performance, in terms of reliability, is impacted by many factors. Examples include basic design, material selection, production processing, installed environment, subsystem or component interfaces, accuracy of predicted load, and maintenance. Therefore, all modes of failure must be anticipated and analyzed.

Reliability data can be obtained from monthly reports produced by AFLC. Examples of suitable data are the RCS:LOG-MMO(AR)-7170 entitled "Maintenance Actions, Manhours, and Aborts by Work Unit Code," which provides historical information pertaining to each assigned WUC. You have to identify the air vehicle and appropriate work unit codes for the pneumatic power system. The most pertinent data produced is the monthly MTBF and MTBM for each item, RCS:LOG-MMO(AR)-7220 entitled "Maintainability Reliability Summary," which provides 12 months summary by air vehicle and includes "on" equipment maintenance action occurrences and manhours and "off" equipment units and manhours as reported on AFTO Form 349. These reports are obtainable from AFLC/MMOMA upon request.

REQUIRMENT LESSONS LEARNED

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4.3 Verification of reliability. Compliance with specified reliability requirements shall be verified by _____.

VERIFICATION RATIONALE (4.3)

Reliability data is needed for overall mission planning, and resource allocations.

VERIFICATION GUIDANCE

Most pneumatic system components are tested during the development program. Individual components are exposed to qualification tests to system contractor specifications. Various components may be tested in the system simulator and are performing on the air vehicle during various phases of flight testing. Ask the reliability monitor for assistance.

VERIFICATION LESSONS LEARNED

3.4 Maintainability. The pneumatic power system maintainability requirements shall be integrated with the overall system maintainability programs and shall comply with the requirements of MIL-STD-470. Special maintainability requirements are as follows: _____.

REQUIREMENT RATIONALE (3.4)

Pneumatic power system maintainability is an integral part of the maintainability requirements for the total air vehicle. Therefore, limits on maintainability tasks and the associated times to accomplish these tasks should be generated for the pneumatic power system and its components. In order to achieve best overall system maintainability, the allotted tasks must be individually considered.

REQUIREMENT GUIDANCE

The overall system maintainability monitor should assist in establishing these requirements. For details relating to maintainability programs consult MIL-STD-470 and MIL-STD-471.

Component or subsystem performance in terms of maintainability is influenced and impacted by many factors, including complexity of design, intended SE, design life, servicing requirements, and bulk of the hardware.

Maintainability performance data can be obtained from monthly reports produced by AFLC. Examples include: RCS:LOG-MMO(R)-7170 entitled "Maintenance Actions, Manhours, and Aborts by Work Unit Code." This report produces monthly MTBF and MTBM for each component or WUC. Another report is RCS:LOG-MMO(AR)-7220 entitled "Maintainability Reliability Summary," which provides 12 months summary by air vehicle and includes "on" equipment maintenance action, occurrences, and manhours. It also includes "off" equipment and manhours as reported on AFTO Form 349. These reports are obtainable from AFLC/MMOMA upon request.

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Components that require frequent removal should be attached to the primary structure with easily removable hardware.

Pneumatic power system maintainability is significantly enhanced by designing for maximum interchangeability of components.

Components should be designed for adjustment and repair at the lowest level of maintenance as far as practical. Adequate technical data is vital to efficient performance of all required adjustments and repairs. Technical data distribution must be consistent with the intended level of repair.

Avoid adjustments or repairs on the air vehicle if such actions are difficult to accomplish due to access or environment. Consider use of replaceable modules or components that can be adjusted or repaired in a field shop as an alternative.

REQUIREMENT LESSONS LEARNED

Consider a bleed fitting to reduce main manifold pressure for maintenance or system check such as compressor cut in and cut out, moisture separator blowdown, etc. Subsystem operation to accomplish this is hard on system components, e.g. cycling F3H RAT to bleed-down system resulted in fatigue failures.

4.4 Verification of maintainability. Compliance with specified maintainability requirements shall be verified by _____.

VERIFICATION RATIONALE (4.4)

Maintainability data is needed to generate meaningful technical data and to verify required skill levels.

VERIFICATION GUIDANCE

Maintenance of pneumatic power system equipment is accomplished at various levels. The task allocation should reflect the overall logistic plan and be consistent with the intended inventory practices. Accessibility to the pneumatic power system components subject to frequent removals should be verified using tools normally found in a mechanic's tool box. Ask the maintainability monitor for assistance.

VERIFICATION LESSONS LEARNED

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3.4.1 Service and access. All pneumatic power system components shall be accessible for inspection, cleaning, adjustment, or replacement, with tools normally found in a mechanic's tool box, while installed and without removal of the engine or major parts of the structure. Exceptions shall be identified.

REQUIREMENT RATIONALE (3.4.1)

Good access to all pneumatic power system components is necessary to permit inspection, adjustment, and repair and/or replacement.

REQUIREMENT GUIDANCE

Access doors and modular construction should be considered when "open" access is not available.

REQUIREMENT LESSONS LEARNED

4.4.1 Verification of service and access verification. Service and access shall be verified by inspection and demonstration.

VERIFICATION RATIONALE (4.4.1)

The best method to verify service and access is by demonstration in a working environment.

VERIFICATION GUIDANCE

Service and access should be evaluated on a mockup prior to, and during, full-scale development.

VERIFICATION LESSONS LEARNED

3.5 System safety. The pneumatic power system safety requirements shall comply with the requirements of _____.

REQUIREMENT RATIONALE (3.5)

System safety requirements for the pneumatic power system should reflect the requirements for the total air vehicle.

REQUIREMENT GUIDANCE

The subsystem and component designs must take into consideration the hazards generated by various modes of failure and ensure means to avoid those modes which potentially can create Category I, Catastrophic, and Category II,

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Critical hazards per MIL-STD-882. Category III, Marginal, and Category IV, Negligible hazards, although qualitatively of less consequence than Category I or Category II hazards, may impact the design because of their anticipated frequency of occurrence. The safety analysis and design change process involves identification of all potential hazards, hazard reduction and minimization techniques, and a closed-loop procedure established to assure design change, as appropriate. System safety for the pneumatic power systems ties directly to MIL-STD-882 and the System Safety Program Plan.

REQUIREMENT LESSONS LEARNED

4.5 Verification of system safety. Compliance with specified system safety requirements shall be verified by _____.

VERIFICATION RATIONALE (4.5)

Safety criteria is needed to assess first flight readiness and to identify potential hazards.

VERIFICATION GUIDANCE

System safety assessment is an integral part of the system safety program for the total air vehicle. Safety is generally verified by analytical methods, which involve preliminary hazard analysis, subsystem hazard analysis, system hazard analysis, and operating hazard analysis. Fault hazard analysis is an inductive tool and fault tree analysis is a deductive tool to identify hazards. Finally, recognition should be made of the fact that if hazards, especially catastrophic or critical, cannot be controlled to an acceptable level, then ground or flight test should include measures to verify the safety of the pneumatic power system.

Prior to the first flight, the following pneumatic power system safety assessment criteria should be accomplished:

a. Establish that all components are proven safe by suitable test or experience.

b. Establish that safe subsystem operation is confirmed by analysis and appropriate tests.

c. Establish that normal and emergency modes of operation of the system are confirmed by test.

d. Establish that all subsystem limitations are defined.

e. Establish that suitable normal and emergency procedures are included in the flight manual.

f. Establish that component installation is in accordance with established standards.

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VERIFICATION LESSONS LEARNED

3.6 Interface requirements. Pneumatic power system interfaces shall be identified and controlled to ensure form, fit, and functional compatibility.

REQUIREMENT RATIONALE (3.6)

Interface baseline design data should be established in the aircraft development phase to minimize the need for extensive design changes which emerge during aircraft assembly and test.

REQUIREMENT GUIDANCE

Pneumatic power system interfaces should be identified in pneumatic power system design reports and appropriate contractual documents.

REQUIREMENT LESSONS LEARNED

4.6 Verification of interface requirements. Pneumatic power system interface(s) with other air vehicle systems shall be verified by _____.

VERIFICATION RATIONALE (4.6)

Interface control data is necessary to ensure physical and functional compatibility.

VERIFICATION GUIDANCE

Verification may be accomplished by analysis, demonstration, inspection, and test.

VERIFICATION LESSONS LEARNED

3.6.1 Electrical power system interface. All electrical components shall operate with electrical power defined by _____.

REQUIREMENT RATIONALE (3.6.1)

The quality and quantity of available power should be matched to the required needs of the system and component operation(s).

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REQUIREMENT GUIDANCE

The electrical power available should be defined at the system and component level. Refer to MIL-STD-704 or the Airborne Electrical Power Specification for guidance. The number of different electrical connectors should be identified and limited to reduce life-cycle costs.

REQUIREMENT LESSONS LEARNED

There have been cases where D.C. solenoids would not overcome component forces at minimum voltage.

4.6.1 Verification of electrical power system interface. Electrical power system interface shall be verified by _____.

VERIFICATION RATIONALE (4.6.1)

Physical and functional compatibility with electrical system(s) are paramount for pneumatic system performance.

VERIFICATION GUIDANCE

Operation of pneumatic power system components at specified power should be verified by system and component tests. Required performance should be demonstrated for all variations permitted by the specification (i.e. at both ends of the allowable voltage range).

VERIFICATION LESSONS LEARNED

3.6.2 Support equipment (SE) interface. The pneumatic power system shall be compatible with the following SE:

a. _____.

b. _____.

c. _____.

REQUIREMENT RATIONALE (3.6.2)

All SE which is in operation and is expected to be compatible with new aircraft systems should be identified to insure interface compatibility.

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REQUIREMENT GUIDANCE

Identify SE such as pneumatic servicing cart, nitrogen servicing equipment, et cetera. Ground service connections for pneumatic system checkout should be accessible at ground level for maintenance personnel and so specified.

Further identify peculiar fittings to mate with the SE.

REQUIREMENT LESSONS LEARNED

4.6.2 Verification of support equipment (SE) interface. The interface of the pneumatic power system with specified support equipment shall be verified by _____.

VERIFICATION RATIONALE (4.6.2)

Physical and functional interface of support equipment is essential for timely checkout and maintenance.

VERIFICATION GUIDANCE

Compatibility of SE should be verified by ground demonstration, analysis, and inspection.

VERIFICATION LESSONS LEARNED

3.6.3 Instrumentation interface(s). Instrumentation interface(s) shall be identified.

REQUIREMENT RATIONALE (3.6.3)

The type and quantity of instrumentation needed for a subsystem is dependent upon the desires of the operator and the complexity and flexibility of the subsystem.

REQUIREMENT GUIDANCE

As a minimum the following instrumentation should be considered:

a. Pneumatic power system pressure (each system)

b. Low-Pressure Indicator (each system)

A simple system requires limited instrumentation. A system with little flexibility or few alternate modes of operation requires only limited instrumentation since, if a malfunction is detected, corrective action is not possible.

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The procuring activity preference for instrumentation should be stated. For redundant instrumentation systems, operator warning should be considered when the systems pressures do not agree within a specified tolerance.

REQUIREMENT LESSONS LEARNED

4.6.3 Verification of instrumentation interface(s). The presence of required instrumentation shall be verified by _____. The accuracy and range of the instrumentation shall be verified by _____.

VERIFICATION RATIONALE (4.6.3)

The physical and functional interface of instrumentation systems are essential to measure the health of the pneumatic system.

VERIFICATION GUIDANCE

The presence of the required instrumentation should be verified by inspection.

The accuracy and range of the instrumentation should be verified by component testing and by installed system calibration.

VERIFICATION LESSONS LEARNED

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a. Paragraph Number and Wording:

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